Digital Signal Processing

Zhou Hang

Optimal design of linear phase FIR filter

- Basic principle of optimal design for linear phase FIR DF
- Design FIR DF using the integral weighted squared error criterion
- Design FIR DF using the minimax error criterion

The basic principle of optimal design

➢The concept of optimal design:

Under a certain error criterion, minimize the error ε between the frequency response $H(e^{j\Omega})$ of the designed FIR filter and the frequency response $H_d(e^{j\Omega})$ of the given filter.

The basic principle of optimal design

➢Common error criteria:

1. Minimax error criterion(Chebyshev criterion)

$$\varepsilon = \max_{\Omega \subset I} \left| W(\Omega)[A(\Omega) - D(\Omega)] \right|$$

The weighted error function can be represented as $E(\Omega)$

$W(\Omega) \geq 0$, is weighting function defined by the designer

Magnitude function of the designed filter

Magnitude function of the given filter

$A(\Omega)$ should have the amplitude function form of a linear phase filter of Type I, II, III, or IV

I: The collection of frequency bands within the range $0 \leq \Omega \leq \pi$ for a filter, such as a low-pass filter $I=[0,\Omega_p] \cup [\Omega_s, \pi]$。

The basic principle of optimal design

➢Common error criteria:

2. Integral weighted squared error criterion

$$\varepsilon = \int_I \left[W(\Omega)[A(\Omega) - D(\Omega)] \right]^2 \mathrm{d}\Omega$$

The basic principle of optimal design

The amplitude function forms of the four types of linear phase filters

Type I $$A(\Omega) = h[M/2] + \sum_{k=0}^{M/2} 2h[M/2-k]\cos(k\Omega)$$

Type II $$A(\Omega) = \sum_{k=0}^{(M-1)/2} 2h[M/2-k]\cos[(k+0.5)\Omega]$$

Type III $$A(\Omega) = \sum_{k=0}^{M/2} 2h[M/2-k]\sin(k\Omega)$$

Type IV $$A(\Omega) = \sum_{k=0}^{(M-1)/2} 2h[(M-1)/2-k]\sin[(k+0.5)\Omega]$$

The four types of linear phase filters can be uniformly represented as $A(\Omega) = Q(\Omega)G(\Omega)$

The basic principle of optimal design

➢Basic principle: $A(\Omega) = Q(\Omega)G(\Omega)$

Based on a certain error criterion, the calculation of $g[k]$ aims to minimize the error ε within the range $0 \le \Omega \le \pi$ defined by that error criterion. The coefficients $h[k]$ of the FIR filter are determined by $g[k]$.

For the integral weighted squared error criterion, the objective is to calculate the coefficients $g[k]$ of the FIR filter in such a way that the error $\varepsilon = \int_I \left[E(\Omega)\right]^2 \mathrm{d}\Omega$ is minimized, and then to use $g[k]$ to determine $h[k]$.

For the minimax error criterion, the objective is to calculate the coefficients $g[k]$ of the FIR filter in such a way that the error $\varepsilon = \max_{\Omega \subset I} \left|E(\Omega)\right|$ is minimized, and then to use $g[k]$ to determine $h[k]$.

Design FIR DF using the integral weighted squared error criterion

According to the integral weighted squared error criterion, Designing a filter involves calculating the coefficients $g[k]$ such that the error $\varepsilon = \int_I \left[W(\Omega)[Q(\Omega)G(\Omega) - D(\Omega)] \right]^2 \mathrm{d}\Omega$ is minimized, and then using $g[k]$ to determine $h[k]$.

Taking the derivative of ε with respect to $g[k]$ and setting it equal to 0 yields the coefficients $g[k]$ that minimize the error.

This method can be implemented using the function firls provided by MATLAB

Design FIR DF using the minimax error criterion

How does the minimax criterion obtain the filter coefficients $h[k]$ that correspond to the minimum error ε?

PM (Parks-McClellan) Algorithm or Remez Algorithm

James McClellan and Thomas Parks proposed an iterative algorithm in 1972 for finding the optimal FIR digital filter.

During the solution process, the Remez exchange algorithm from mathematical optimization is utilized. Therefore, this method is known as the PM (Parks-McClellan) algorithm or simply the Remez algorithm.

[M,fo,ao,w] = firpmord (f,a,dev) Estimate the filter order M

h = firpm (M,fo,ao,w) Implementing the PM algorithm for digital filter design

Design FIR DF using the integral weighted squared error criterion



Lancaster University College at Beijing Jiaotong University

MATLAB Implementation $h = firls(M,f,a,w)$

M: Filter Order

f: Normalized band values. For example, the B bands of an FIR filter are

$$\pi f_1 \le \Omega \le \pi f_2, \quad \pi f_3 \le \Omega \le \pi f_4, \quad \ldots, \quad \pi f_{2B-1} \le \Omega \le \pi f_{2B}$$

Then f is a vector with 2B elements, and its value is

$$f=[f_1, f_2, \ldots, f_{2B-1}, f_{2B}]$$

a: Amplitude Values It represents the amplitude of the filter at the boundaries of the aforementioned frequency bands.

w: The weighted values for each frequency band Setting different minimum error values for different frequency segments within those bands.

For example, design a linear phase FIR low-pass digital filter with M=63（Type II）, Ω_p=0.5π rad, and Ω_s=0.6π rad, based on the integral weighted squared error criterion.

Analyze: Using the MATLAB function firls to implement, the call format is

h = firls(M,f,a,w)

Call parameter:

M=63; % Order

f=[0 Fp Fs 1]; %a low-pass filter frequency band (passband frequency Fp=0.5, Stopband frequency Fs=0.6)

a=[1 1 0 0] ; % The amplitude of the filter at the frequency band boundaries in f

w the weighting values w for each frequency band are the same, they can be omitted.

That is to say, h=firls(M,[0 Fp Fs 1],[1 1 0 0]);

For example, design a linear phase FIR low-pass digital filter with M=63（Type II）, Ω_p=0.5π rad, and Ω_s=0.6π rad, based on the integral weighted squared error criterion.

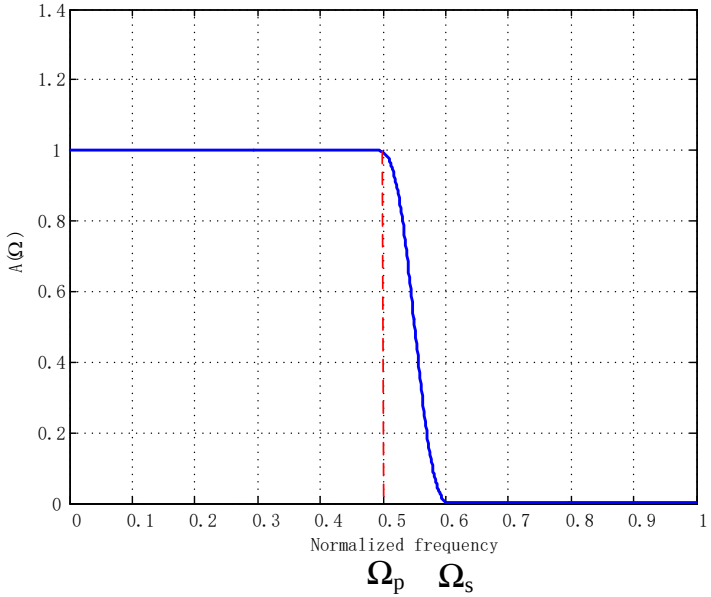
Solution： The MATLAB program is as follows

```
% Design an FIR filter based on the integral weighted squared error criterion

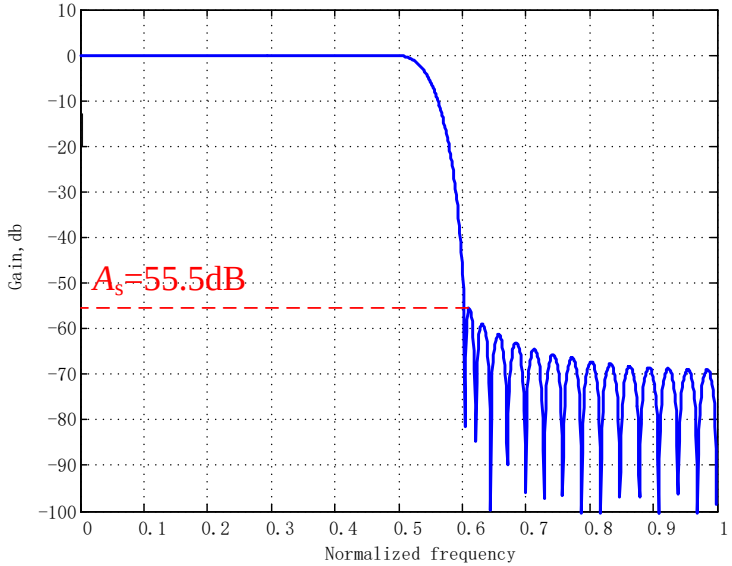
M=63;Fp=0.5;Fs=0.6;  % Define filter parameters
h=firls(M,[0 Fp Fs 1],[1 1 0 0]);  %Low-pass filter
w=linspace(0,pi,1000);
mag=freqz(h,[1],w);    %Calculate frequency response
plot(w/pi,abs(mag), 'LineWidth',2);grid on;  % Plot the amplitude response
xlabel('Normalized frequency');ylabel('A(\Omega)');
figure;
plot(w/pi,20*log10(abs(mag)), 'LineWidth',2);grid on;  % Plot the gain response
xlabel('Normalized frequency');ylabel('Gain,db');
axis([0 1 -100 10]);
```

For example, design a linear phase FIR low-pass digital filter with M=63（Type II）, Ω_p=0.5π rad, and Ω_s=0.6π rad, based on the integral weighted squared error criterion.

Solution: Design Result M =63, A_s=55.5dB



Filter amplitude response



Filter gain

Design FIR DF using the minimax error criterion

[M,fo,ao,w] = firpmord (f,a,dev) Estimate the filter order *M*

f: The B bands that need to be designed for the FIR filter

If the B bands for the FIR filter are $0 \le \Omega \le \pi f_1$, $\pi f_2 \le \Omega \le \pi f_3$, …, $\pi f_{2(B-1)} - 2 \le \Omega \le \pi f_{2(B-1)-1}$, $\pi f_{2B-2} \le \Omega \le \pi$

Then f is a vector with 2B-2 elements (B≥2),
and its value is $f=[f_1, f_2, f_3, \ldots, f_{2B-2}]$(Normalized frequency)

a: The amplitude values of the FIR filter in the B bands
have a vector of B elements, where the passbands are set to 1 and the stopbands are set to 0

dev: The fluctuation values of the FIR filter in the B bands have a vector of B elements

M: Filter order

fo,ao: The B bands with 2B boundary frequency values and amplitude values have a vector of B elements

w: The weighted values for each frequency band have a vector of B elements

Design FIR DF using the minimax error criterion

[M,fo,ao,w] = firpmord (f,a,dev) Estimate the filter order *M*

h = firpm (M,fo,ao,w) implement the PM algorithm for digital filter design

M, fo, ao, w: The return value of the function firpmord

h: The unit impulse response $h[k]$ of an M-order linear phase FIR filter

If the designed filter does not meet the design specifications, you can increase the order based on the estimated M, for example, by setting $M = M + 2$.

Example: Design a linear-phase FIR low-pass filter using the PM algorithm that meets the following specifications Ω_p=0.2π rad， Ω_s=0.4π rad, $A_p \le 0.3$dB, $A_s \ge 55$dB



Lancaster University College at Beijing Jiaotong University

Solution: Use the MATLAB functions firpmord and firpm to implement,The call formats are as follows

```
[M,fo,ao,w] = firpmord(f,a,dev)
h =  firpm (M,fo,ao,w)
```

Call parameters：

```
f=[Fp,Fs];% The normalized frequency band for low-pass filtering,
where Fp=0.2,Fs=0.4
a=[1 0] ;% The amplitude values of the filter at the frequency bands defined by f
dev=[δp δs] ;% The fluctuation values of the filter at the frequency bands defined
by f
```

$\delta_p = 1 - 10^{-A_p/20}$ $\delta_s = 10^{-A_s/20}$

Example: Design a linear-phase FIR low-pass filter using the PM algorithm that meets the following specifications Ω_p=0.2π rad， Ω_s=0.4π rad, $A_p \le 0.3$dB, $A_s \ge 55$dB



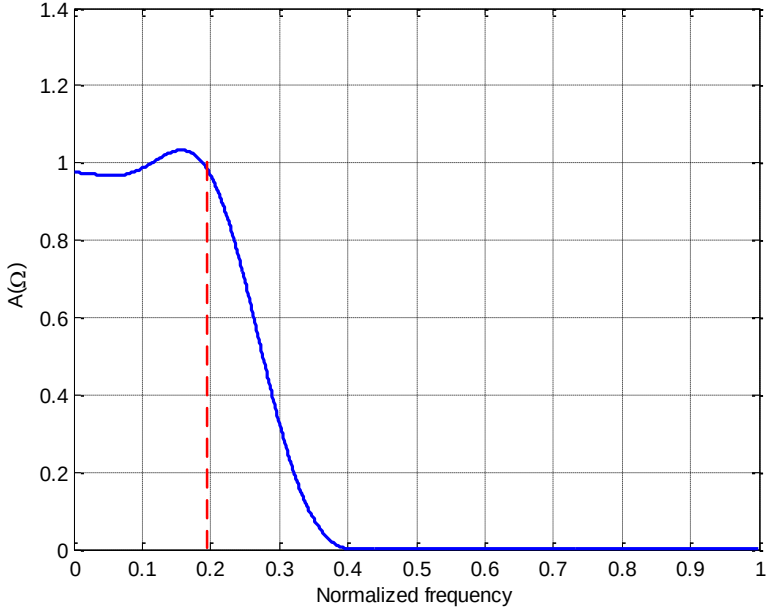
Lancaster University College at Beijing Jiaotong University

Solution:

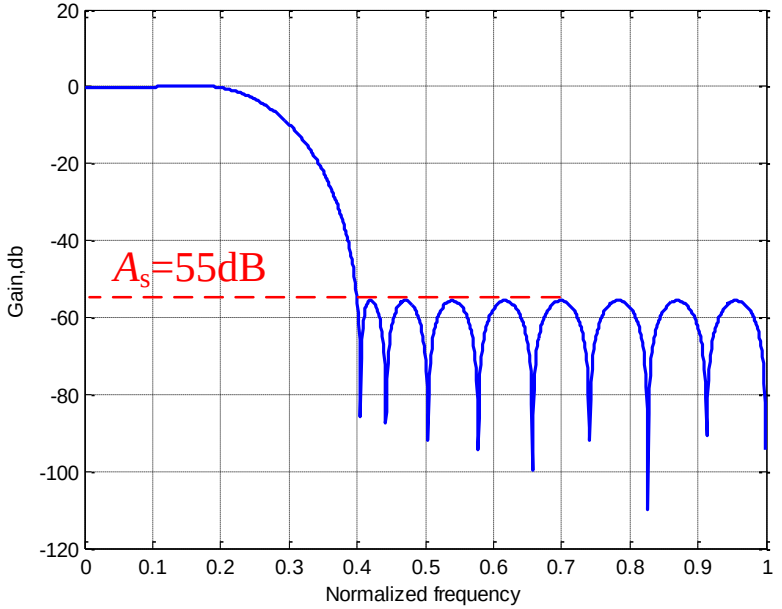
```
%Design an FIR filter using the PM algorithm
Fp=0.2;Fs=0.4;ds=0.0017;dp=0.034;
f=[Fp Fs];a=[1 0];dev=[dp ds]; %Determine the filter parameters
[M,fo,ao,w] = firpmord (f,a,dev); % Determine the filter order
M=M+2;%Adjust the filter order
h = firpm (M,fo,ao,w);  % Design a filter using the PM algorithm
w=linspace(0,pi,1000);
mag=freqz(h,[1],w);  % Calculate frequency response
plot(w/pi,abs(mag),'LineWidth',2);grid on;
xlabel('Normalized frequency');ylabel('A(\Omega) ');  % Plot the amplitude response
figure;
plot(w/pi,20*log10(abs(mag)), 'LineWidth',2);grid on;  % Calculate phase response
xlabel('Normalized frequency');ylabel('Gain,db');
```

Example: Design a linear-phase FIR low-pass filter using the PM algorithm that meets the following specifications Ω_p=0.2π rad， Ω_s=0.4π rad, $A_p \le 0.3$dB, $A_s \ge 55$dB

Solution: design results M =21, A_p=0.3dB, A_s=55dB



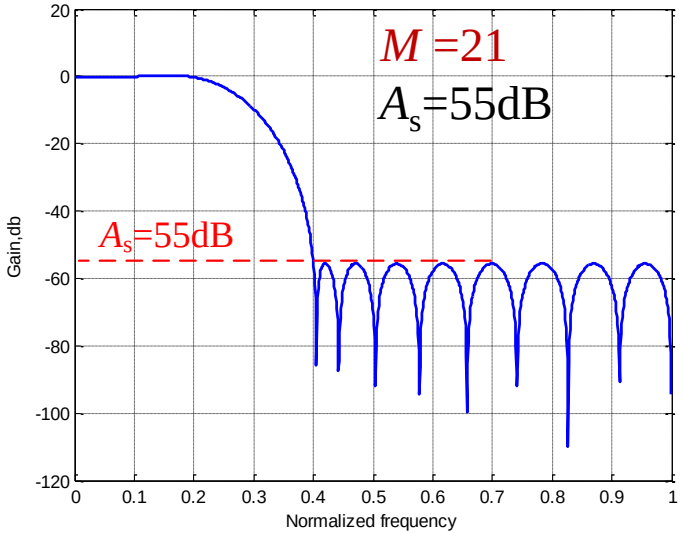
Amplitude response of the filter



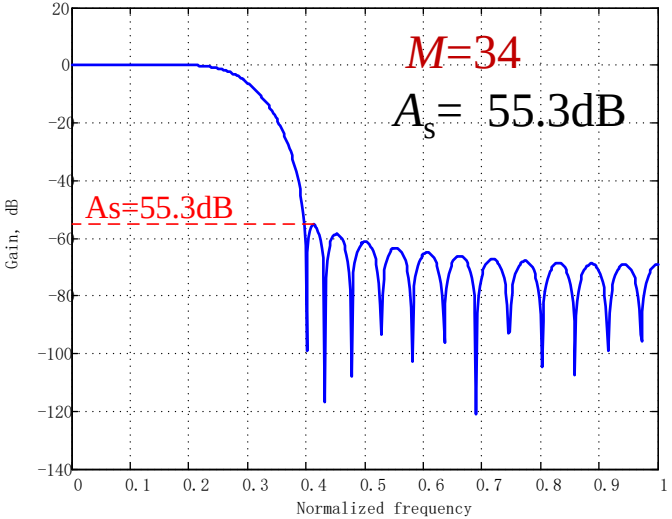
Filter gain

Example: Design a linear-phase FIR low-pass filter using the PM algorithm that meets the following specifications $\Omega_p=0.2\pi$ rad，$\Omega_s=0.4\pi$ rad, $A_p \le 0.3$dB, $A_s \ge 55$dB

Solution: Compare with the design results using a Kaiser window



Gain of the filter designed by PM algorithm

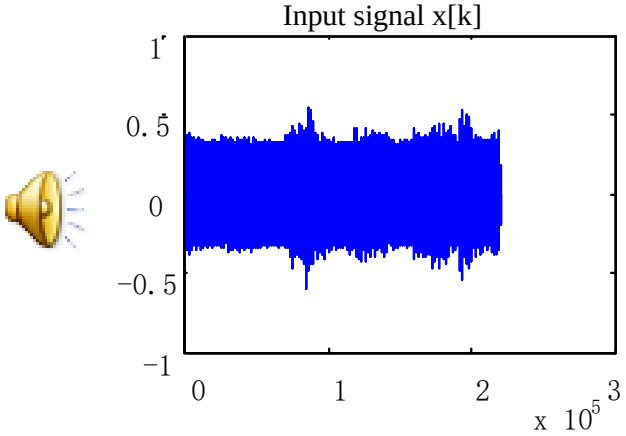


Gain of the filter designed by Kaiser Window

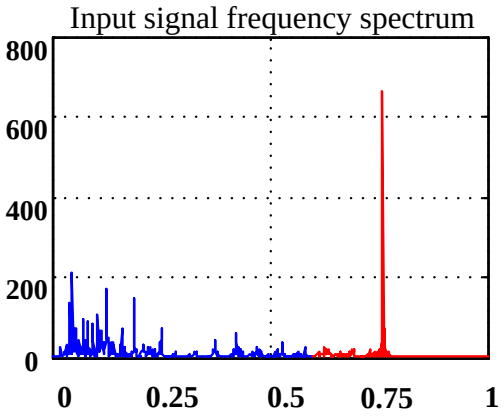
It can be seen that to achieve the same specifications, the order of the equiripple filter is lower than that of the window function method.

Example: Design an FIR DF to remove noise from a given digital audio signal (f_{sam}=11025Hz)

Lancaster University College at Beijing Jiaotong University



Original audio time-domain waveform



Spectrum of Digital Audio

Based on the frequency spectrum, the following design specifications can be determined:

$\Omega_p=0.7\pi$ rad，$\Omega_s=0.75\pi$ rad，$A_p \le 1$dB，$A_s \ge 48$dB

Select the window function method and the PM algorithm to design FIR DF

Example: Design an FIR DF to remove noise from a given digital audio signal (f_{sam}=11025Hz)



Lancaster University College at Beijing Jiaotong University

Solution: Select the Hamming window

```
% Design and Application of FIR filter with hamming window
[x,Fs,nBit]=wavread('C:\yourn.wav'); %Read the audio
N=length(x);
subplot(2,2,1);  plot(x); % Plot the input signal
title('Input signal x[k]');
% Display the spectrum of the input signal
w=linspace(0,pi,1000);
X=freqz(x,[1],w); %Obtain the filter's amplitude frequency
subplot(2,2,2)
plot(w/pi,abs(X)); grid on; ; %Plot the spectrum of the input signal
title('Input signal frequency spectrum');
% Design a low-pass FIR filter
Wp=0.7; Ws=0.75; %Define the filter Parameters
Wc=(Wp+Ws)/2;
N=ceil(7.0/(Ws-Wp)); %Length of the filter N
M=N-1 %OrderM
k=0:M;
h=Wc*sinc(Wc*(k-M/2));
w0=hamming(N);  %Use hamming window
h=h.*w0';
w=linspace(0,pi,1000);
mag=freqz(h,[1],w);  % Calculate frequency response
% Plot the amplitude response of the filter
subplot(2,2,4); plot(w/pi,abs(mag),'r-');  grid on;
title('FIR Filter frequency response');
y=filter(h,[1],x);  % Filtering
Y=freqz(y,[1],w);  % Frequency response characteristics after signal filtering
% Plot the amplitude response of the signal after passing through the filter
subplot(2,2,3);  plot(w/pi,abs(Y));  grid on;
```

Example: Design an FIR DF to remove noise from a given digital audio signal (f_{sam}=11025Hz)



Lancaster University College at Beijing Jiaotong University

Solution: Select the Hamming window

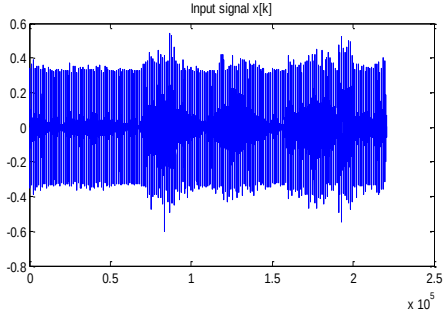
```
title('Output signal frequency spectrum');
wavwrite(y,Fs,nBit, 'C:\your22.wav') ; % Save audio
figure;
% Plot the gain response of the filter
plot(w/pi,20*log10(abs(mag)),'linewidth',3); grid on;
axis([0,1,-100,10])
xlabel('Normalized frequency');  ylabel('Gain, dB');
figure;
thet2=unwrap(angle(mag)); % Calculate phase response
figure;
% Plot the phase response of the filter
plot(w/pi,thet2,'LineWidth',3);grid on;
xlabel('Normalized frequency');ylabel('Phase Frequency');
```

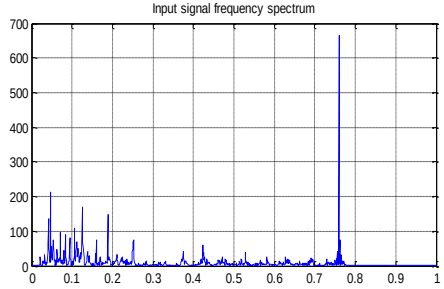
Example: Design an FIR DF to remove noise from a given digital audio signal (f_{sam}=11025Hz)

Lancaster University College at Beijing Jiaotong University

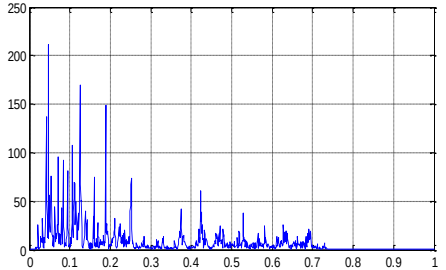
Solution: Select the Hamming window

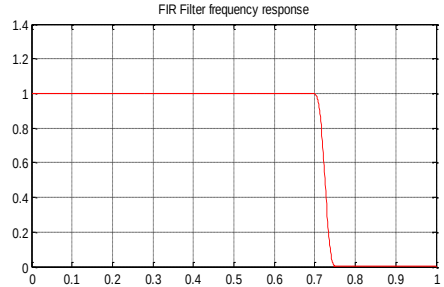












A_p=0.01dB
A_s=54dB
M=139

Example: Design an FIR DF to remove noise from a given digital audio signal (f_{sam}=11025Hz)



Lancaster University College at Beijing Jiaotong University

Solution: Select the PM algorithm to design FIR DF

```
% Design and Application of FIR filter
% Read the audio
[x,Fs,nBit]=wavread('C:\yourn.wav');
N=length(x);
subplot(2,2,1);  plot(x); % Plot the input signaltitle('Input signal x[k]');
% Display the spectrum of the input signal
w=linspace(0,pi,1000);
X=freqz(x,[1],w); % Obtain the filter's amplitude frequency
subplot(2,2,2)
plot(w/pi,abs(X)); grid on; % Plot the spectrum of the input signal
title('Input signal frequency spectrum');
```

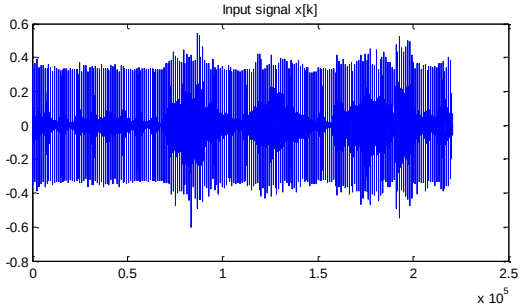
```
% Design a low-pass FIR filter
Fp=0.7;Fs=0.75;ds=0.003;dp=0.1; % Define the filter Parameters
f=[Fp Fs]; a=[1 0]; dev=[dp ds]; % Define as low-pass filtering
[M,fo,ao,w] = firpmord(f,a,dev); % Calculate the order of the filter
h = firpm(M,fo,ao,w);  % Construct the filter
w=linspace(0,pi,1000);
H=freqz(h,[1],w);  % Calculate frequency response
% Plot the amplitude response of the filter subplot(2,2,4);
plot(w/pi,abs(H),'r-');  grid on;
title('FIR Filter frequency response');
y=filter(h,[1],x); % Filtering
Y=freqz(y,[1],w); % Frequency response characteristics after signal filtering
% Plot the amplitude response of the signal after passing through the filter
subplot(2,2,3);  plot(w/pi,abs(Y));  grid on;
title('Output signal frequency spectrum');
```

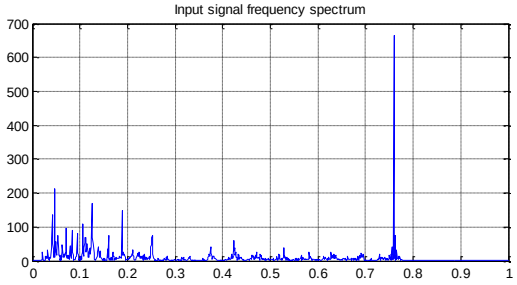
Example: Design an FIR DF to remove noise from a given digital audio signal (f_{sam}=11025Hz)

Lancaster University College at Beijing Jiaotong University

Solution: Select the PM algorithm to design FIR DF

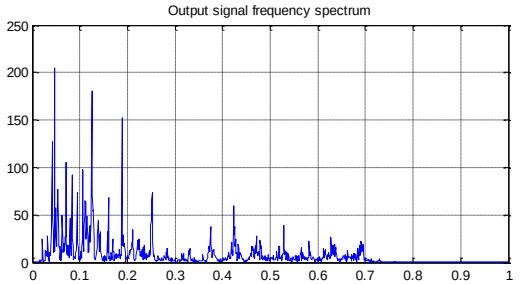


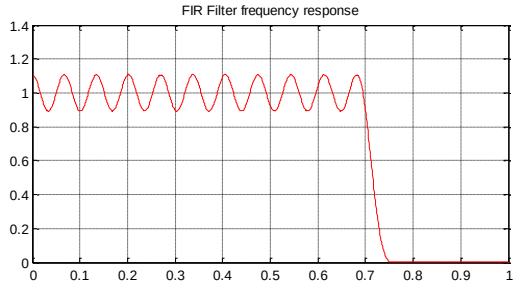




A_p=0.98dB
A_s=50dB
M=61







Compared to the window function method, the filter designed bythe PM algorithm typically has a lower order.

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.